

SMARTDESK AI

AI NOTES:

AI Components Used

- Speech-to-Text Engine (Groq Whisper Large V3)
- Large Language Model (LLaMA 3.1)
- Sentiment Analysis Module
- Employee Information Extraction Module
- Department Routing Engine
- Ticket Classification Engine
- Suggested Resolution Generator
- SQLite Database Management System

Primary AI Features

- Audio Transcription
- Automatic Ticket Generation
- Ticket Title Extraction
- Issue Description Generation
- Category Classification
- Priority Prediction
- Department Assignment
- Sentiment Detection
- Employee Information Extraction
- Ticket Summary Generation
- Suggested Resolution Generation

AI Models Used

- Groq Whisper Large V3 – Speech-to-Text Transcription
- LLaMA 3.1 – Ticket Understanding and Information Extraction
- Regex-Based PII Extractor – Employee Information Detection
- Rule-Based Validation Engine – Output Validation and Correction

Benefits

- Eliminates Manual Ticket Creation
- Reduces Ticket Processing Time
- Improves Ticket Classification Accuracy
- Automates Department Assignment
- Provides Real-Time AI Assistance
- Improves IT Support Efficiency
- Maintains Centralized Ticket Tracking
- Extracts Employee Information Automatically
- Enhances Issue Prioritization
- Supports Faster Resolution Management

Limitations

- Transcription Accuracy Depends on Audio Quality
- Background Noise May Affect Results
- AI Accuracy Depends on User Explanation
- Complex Multi-Issue Calls May Reduce Precision
- Requires Internet Connectivity for Live AI Features
- Requires Valid API Access for AI Processing

Future AI Enhancements

- Real-Time Voice Call Processing
- Duplicate Ticket Detection
- Auto Engineer Assignment
- Multi-Language Support
- SLA Breach Prediction
- Historical Ticket Learning
- AI-Based Resolution Recommendation Engine
- Workload-Based Ticket Distribution
- Advanced Predictive Analytics
- Intelligent Escalation Prediction